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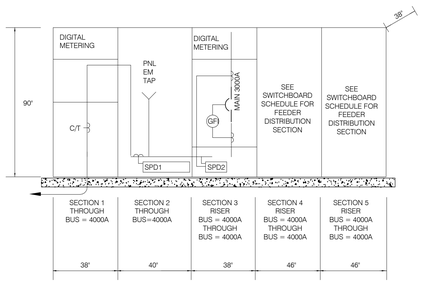


2 E-5.1H SWITCHBOARD MS1 ELEVATION

SCALE: NOT TO SCALE

CONDUCTORS & CONDUITS SCHEDULE

FEEDER SYMBOL	AMPACITY	WIRE SIZE AWG/KCMIL	CONDUIT SIZE
1	3000	COORDINATE WITH BG & E FOR SERVICE ENTRANCE FEEDERS.	-
2	120	5 #1/0 AWG & 1 #6 AWG IG & 1 #6 G	2"
3	20	3 #12 & 1 #12 G.	3/4"
4	405	2 SETS (5 #250 KCMIL & 1 #1/0 AWG IG & 1 #1/0 G)	(2) 3"
5	45	3 #6 & 1 #10 G.	1"
6	110	4 #1 & 1 #6 G.	1 1/2"
7	300	5 #500 KCMIL & 1 #2 AWG IG & 1 #2 G.	3 1/2"
8	400	2 SETS (4 #3/0 & 1 #2 G.)	(2)-2"
9	230	4 #4/0 & 1 #4 G.	2 1/2"
10	255	4 #250 KCMIL & 1 #4 G.	2 1/2"
11	855	3 SETS (4 #300 KCMIL + 1 #2/0 G.)	3-3"
12	175	3 #2/0 & 1 #6 G.	1 1/2"
13	150	4 #1/0 & 1 #6 G.	1 1/2"
14	70	3 #4 & 1 #8 G.	1"
15	350	3 #500 KCMIL & 1 #2 G.	3"
16	55	3 #6 & 1 #6 G.	1"
17	55	4 #6 & 1 #6 G.	1"
18	20	2 #12 & 1 #12 G.	3/4"
19	510	2 SETS (4 #250 KCMIL + 1 #1 G.)	2-2 1/2"

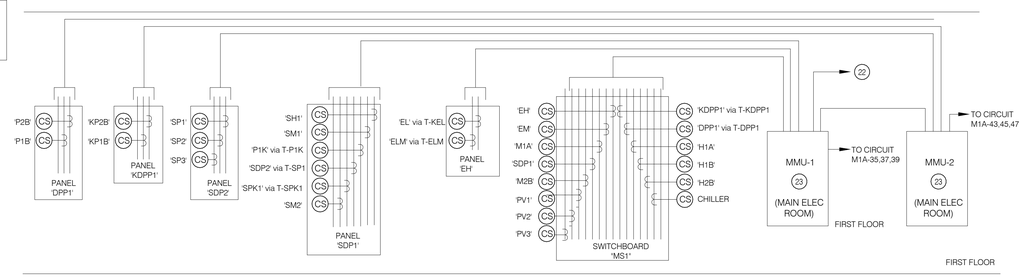
NOTES:
1. PROVIDE CONDUCTORS FOR CONTROLLER OF ALL MECHANICAL EQUIPMENT AS REQUIRED
2. SEE TRANSFORMER SCHEDULE FOR PRIMARY AND SECONDARY FEEDERS SIZE AND GROUNDING ELECTRODE SIZE.
3. PROVIDE LUGS TO MATCH CONDUCTOR SIZES.

KEYED NOTES

1. FEEDERS ENCASED IN CONCRETE. SEE SITE PLAN FOR ADDITIONAL INFORMATION.
2. PERFORM SHORT-CIRCUIT AND COORDINATION STUDY. ALSO PERFORM ARC FLASH HAZARD ANALYSIS IN CONJUNCTION WITH SHORT-CIRCUIT & COORDINATION STUDY. INCLUDES IN THE SCOPE OF WORK PROVIDING ARC FLASH WARNING LABELS. SET THE FIELD ADJUSTABLE CIRCUIT BREAKERS TRIP RANGES. ELECTRICAL CONTRACTOR SHALL HIRE THE MANUFACTURER OF THE BREAKER TO SET TIME CURRENT CURVES ON THE MAIN CIRCUIT BREAKER AND ASSOCIATED BRANCH CIRCUIT BREAKER FOR THE SWITCHBOARD SO THAT THEY ARE SELECTIVELY COORDINATED WITH EACH OTHER. THE INTERRUPTING RATING OF ALL ELECTRICAL EQUIPMENTS SHALL BE BASED ON THE DATA RESULTS FROM THE SHORT-CIRCUIT & COORDINATION STUDY. PROVIDE INSULATED CASE POWER CIRCUIT BREAKER FOR MAIN CIRCUIT BREAKER AND LARGER FRAME SIZE CIRCUIT MOLDED CIRCUIT BREAKER FOR FEEDER BRANCH CIRCUIT BREAKER.
3. CONNECT TO COLD METAL WATER PIPE OR TO THE NEAREST BUILDING STRUCTURAL STEEL. SEE TRANSFORMER SCHEDULE FOR GROUNDING ELECTRODE SIZE.
4. PROVIDE A TOTALIZING KWH DEMAND METER. COORDINATE WITH UTILITY COMPANY FOR HOUSE METER & FIRE PUMP METER.
5. 100A, 3-POLE + S/N, 600V, U.L. LABEL SERVICE ENTRANCE SWITCH FUSED AT 100A.
6. 6" HIGH CONCRETE HOUSE KEEPING PAD.
7. 225 KVA, 480 DELTA-208Y/120V, 3PH, K-FACTOR 13, DRY TYPE TRANSFORMER.
8. PROVIDE IDENTIFICATION AS INDICATED FOR SERVICE EQUIPMENT DISCONNECT PER NEC-2014 ARTICLE 230.(E).
9. ENGINE START CONTACTS AND ASSOCIATED CONTROL WIRING IN 1" CONDUIT TO GENERATOR SET CONTROLLER.
10. 4" HIGH CONCRETE HOUSEKEEPING PAD.
11. SEE DETAIL ON DRAWING E7.1 FOR GROUNDING INFORMATION.
12. 45 KVA, 480 DELTA-208/120V, 3PH, DRY TYPE TRANSFORMER.
13. 3 KVA, 480 DELTA-208Y/120V, 1PH, DRY TYPE TRANSFORMER.
14. 150 KVA, 480 DELTA-208Y/120V, 3PH, DRY TYPE TRANSFORMER.
15. 75 KVA, 480 DELTA-208Y/120V, 3PH, DRY TYPE TRANSFORMER.
16. 260A, 480V, 3-POLE + S/N, 42KAIC, AUTOMATIC TRANSFER SWITCH (ATS) EQUIPPED WITH INTEGRAL SPD.
17. UL LISTED SERVICE ENTRANCE FIRE PUMP DISC. SWITCH (LOCKED ON POSITION) RATED AT 200A, 3-POLE + S/N,600V, FUSED AT 150A. EXACT FRAME AND FUSE SIZE MAY CHANGE TO MATCH FINAL FIRE PUMP SIZE.
18. 800A, 480V, 3-POLE + S/N, 65KAIC, MANUAL TRANSFER SWITCH (MTS) EQUIPPED WITH INTEGRAL SPD
19. 200 KW, 480V, 3PH, 4W, DIESEL ENGINE EMERGENCY/STANDBY GENERATOR EQUIPPED WITH BUILT-IN MAIN CIRCUIT BREAKERS AND SKID MOUNTED FUEL TANK SUFFICIENT FOR 12 HOURS OPERATION AT FULL LOAD (161 GALLONS) IN LEVEL 2 SOUND ATTENUATED WEATHERPROOF ENCLOSURE. PROVIDE BARRIER TO SEPARATE THE TWO BREAKERS. BREAKERS SHALL BE LOCKABLE IN THE OPEN POSITION PER NEC ARTICLE 445.18.
20. THE ON-SITE GENERATOR IS NOT A SEPARATELY DERIVED SYSTEM, AS THE NEUTRAL IS SOLIDLY INTERCONNECTED TO A SERVICE SUPPLIED SYSTEM NEUTRAL.
21. ENGINE START CONTACTS AND ASSOCIATED CONTROL WIRING IN 1" CONDUIT TO MOBILE GENERATOR TERMINATION BOX.
22. ALL MMU's ARE LINKED VIA RS-485 DAISY-CHAINED CONNECTION USING Bacnet MS TP PROTOCOL TO BAS SYSTEM. PROVIDE INTEGRATOR AS REQUIRED FOR COMMUNICATION WITH BAS. COORDINATE SYSTEM CONFIGURATION WITH OWNER.
23. PROVIDE E-MON SUB. METER WITH THE FOLLOWING SPECIFICATION:
    - METER SHALL BE A MMU (MULTIPLE METER UNIT) AND FULLY ELECTRONIC WITH DIGITAL LCD DISPLAY FOR KILOWATT HOUR READINGS, KILOWATT DEMAND, KILOWATT PEAK DATE AND TIME.
    - METER SHALL BE UL LISTED. CSA APPROVED AND CERTIFIED BY A NATIONALLY RECOGNIZED INDEPENDENT TESTING.
    - METER SHALL PROVIDE A LOAD INDICATOR TO INDICATE REAL TIME POWER CONSUMPTION LEVELS FOR FIELD TESTING AND CERTIFICATION.
    - METER SHALL BE PROVIDED WITH A SELF CONTAINED BACK-UP SYSTEM TO MAINTAIN MEMORY DURING POWER FAILURES. THE DATA CAN BE STORED IN THE MEMORY UP TO A YEAR ON UNTIL RESET.
    - METER SHALL BE PROVIDED WITH A MODULAR CONNECTOR TO PROVIDE INTERFACING FOR AUTOMATIC METER READING, PULSE MODULES, ANALOG SIGNAL MODULES, ENERGY CONTROL MODULES, INSTANTANEOUS DEMAND DISPLAYS.
    - METER SHALL BE EQUIPPED WITH Bacnet MS TP PROTOCOL.
    - METER SHALL BE MMU 24 WITH EMON CLASS 3200 METERS WITH BACNET MS TP.
24. 112.5 KVA, 480 DELTA-208Y/120V, 3PH, K-FACTOR 13, DRY TYPE TRANSFORMER.
25. 30 KVA, 480 DELTA-208Y/120V, 3PH, K-FACTOR 13, DRY TYPE TRANSFORMER.
26. PROVIDE A SIGN TO INDICATE TYPE AND LOCATION OF ON-SITE EMERGENCY POWER SOURCES PER NEC ARTICLE 700.7 (A).
27. COMBINATION 100A ATS / FIRE PUMP CONTROLLER FURNISHED BY PLUMBING/FIRE PROTECTION CONTRACTOR AND WIRED BY ELECTRICAL CONTRACTOR.
28. FINAL SIZING FOR ELECTRICAL OVER CURRENT PROTECTION DEVICE AND WIRE SIZE SHALL BE BASED ON FINAL ELEVATOR SELECTION.
29. ELEVATOR SHUNT TRIP FUSED DISCONNECT SWITCH. MANUFACTURED BY "FERRAZ SHAWMUT" OR APPROVED EQUAL. 60A, 3-POLE, 600V WITH AUXILIARY AND ALARM CONTACT FUSED AT 60A. COORDINATE WITH ELEVATOR CONTRACTOR FOR FINAL ELEVATOR MOTOR SIZE AND POWER CONNECTION PRIOR TO INSTALLATION. DISCONNECT SWITCH SHALL BE CAPABLE OF BEING LOCKED IN THE OPEN POSITION PER NEC ARTICLE 620.51(A). PROVIDE BATTERY LOWERING DEVICE.
30. FEEDERS IN 2" ENCASED IN CONCRETE.
31. 30A,1-POLE +S/N, 240V FUSED DISCONNECT SWITCH FUSED AT 20A FOR ELEVATOR CAB LIGHTS AND FAN.
32. PROVIDE 2 HOUR RATED FEEDERS (MINERAL INSULATED CABLE OR ENCASED IN CONCRETE.
33. REFER TO "W" SERIES DRAWINGS FOR PHOTOVOLTAIC DESIGN.

KEYED NOTES

34. 15 KVA, 480 DELTA-208Y/120V, 3PH, DRY TYPE TRANSFORMER. REFER TO "W" SERIES DRAWINGS FOR TRANSFORMER TYPE AND EXACT LOCATION.



CS E-MON SPLIT-CORE CURRENT SENSOR

MMU-1: E-MON MMU-24 MULTI-METER UNIT

MMU-2: E-MON MMU-8 MULTI-METER UNIT

3 E-5.1H ENERGY MONITORING DIAGRAM

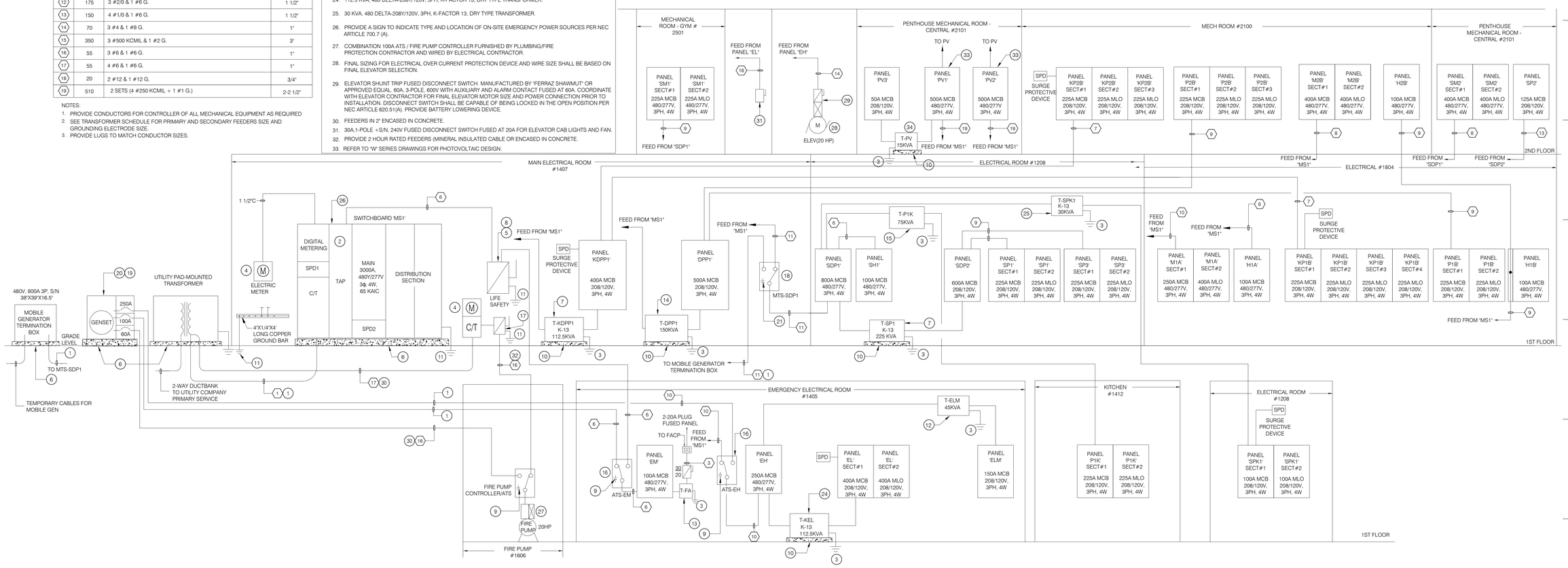
SCALE: NOT TO SCALE

DRY TYPE TRANSFORMER SCHEDULE: 480 DELTA-208Y/120V, 3-PHASE

TRANSFORMER ID	RATING (kVA)	PRIMARY				SECONDARY				
		FLA (AMPS)	MAX OCP	FEEDER SIZE	CONDUIT SIZE	FLA (AMPS)	MAX OCP	FEEDER SIZE	CONDUIT SIZE	GROUNDING ELECTRODE
T-SPK1	30	36.1	45	3 #6 AWG + 1 #10 AWG G	1"	83.3	100	5 #1/0 AWG + 1 #6 AWG IG + 1#6 G	2"	#6
T-ELM	45	54.1	70	3 #4 AWG + 1 #8 AWG G	1"	124.9	150	4 #1/0 KCMIL + 1 #6 AWG G	1 1/2"	#6
T-SP1	225	270.6	350	3 #500 KCMIL + 1 #2 AWG G	3"	624.5	600	2 SETS(5 # 500 KCMIL + 1 # 1/0 AWG IG + 1 # 1/0 AWG G)	2-4"	#1/0
T-P1K	75	90.2	110	3 #1 AWG + 1 #6 AWG G	1 1/4"	208.2	250	4 #250 KCMIL + 1 #2 AWG G	2 1/2"	#2
T-KEL	112.5	135.3	175	3 #2/0 AWG + 1 #6 AWG G	1 1/2"	312.3	400	2 SETS(5 # 250 KCMIL + 1 # 1/0 AWG IG + 1 # 1/0 AWG G)	2-3"	#1/0
T-KDPP1	112.5	135.3	175	3 #2/0 AWG + 1 #6 AWG G	1 1/2"	312.3	400	2 SETS(5 # 250 KCMIL + 1 # 1/0 AWG IG + 1 # 1/0 AWG G)	2-3"	#1/0
T-DPP1	150	180.4	225	3 #4/0 AWG + 1 #4 AWG G	2"	416.4	500	2 SETS (4 #250 KCMIL + 1 #1/0 AWG G)	2-2 1/2"	#1/0
T-PV	15	18	30	3 #10 AWG + 1 #10 AWG G	3/4"	41.6	50	4 #6 AWG + 1 #8 AWG G	1"	#8

DRY TYPE TRANSFORMER SCHEDULE: 480 DELTA-208Y/120V, 1-PHASE

TRANSFORMER ID	RATING (kVA)	PRIMARY				SECONDARY				
		FLA (AMPS)	MAX OCP	FEEDER SIZE	CONDUIT SIZE	FLA (AMPS)	MAX OCP	FEEDER SIZE	CONDUIT SIZE	GROUNDING ELECTRODE
T-FA	3	6.3	20	2 #10 AWG + 1 #10 AWG G	3/4"	14.4	20	3 #8 AWG + 1 #8 AWG G	3/4"	#8



1 E-5.1H ELECTRICAL POWER RISER DIAGRAM-H

SCALE: NOT TO SCALE

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ELECTRICAL POWER RISER DIAGRAM

HOLABIRD ACADEMY PK-8
CITY OF BALTIMORE, MARYLAND

DATE	DESCRIPTION

E-5.1H

03/13/2017
BID SET

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